Functional Consciousness: A Proxy Metric Using Self-Models

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Abstract. This paper proposes Functional Consciousness (FC) as a measurable architectural property: the observable capacity of a system to access and reason about internal representations of its own states. We introduce a computationally tractable metric based on self-models and their associated reasoning power, measured through informational richness and state-space expansion under inference. The resulting Functional Consciousness Score (FCS) is applied to benchmark systems with known internal structure, including a Waymo L4 autonomous vehicle. To extend the framework to black-box systems, we present Functional Self-Model Analysis (FSMA), an abductive methodology for inferring self-models from behavioral evidence. Applied to stream-of-consciousness literature, FSMA yields a catalog of self-models that serves as a reference for estimating functional consciousness in more complex biological and artificial agents. The resulting scores align with intuitive gradients of cognitive sophistication while remaining operationally grounded. Finally, we compare FC with major theories of consciousness and argue that several of their central functional claims become partially measurable within this framework.

Keywords: AI Benchmarking · AGI · Cognitive Architectures · Consciousness · IIT · Information Theory · Machine Consciousness · Philosophy of Mind

Introduction

An AI system cannot be genuinely "general" if it cannot become an object of its own reasoning. As systems move from narrow task execution toward autonomous, general behavior, the ability to represent, inspect, and reason about internal states becomes a central engineering problem. This capacity is required for AI safety, reasoning about an agent's own cognitive limits, and Theory of Mind (modelling others like yourself) [19]. It also addresses practical issues like memory reorganization and error reporting. However, it remains unclear how to define or measure this architectural feature.

This paper proposes "Functional Consciousness (FC)" to capture a system's observable capacity to access and reason about its internal states. We deliberately

sidestep the "hard problem" of consciousness [7] and "phenomenal consciousness"—the question of "what it feels like" to a subject.

Instead, we ground our work in functionalism [20] and access consciousness [6]. These frameworks define mental states by their functional roles in producing observable behavior. Viewed this way, Functional Consciousness becomes a tangible architectural property rather than an elusive quality.

Definition 1 (Functional Consciousness, FC). The observable capacity of a system to access and process internal representations of its own states to produce behavior.

By adding the "self-model" as a unit of analysis, we can apply the tools of information theory [5] to measure the predictive and reasoning power of these models. In this section, we validate this white-box metric by benchmarking agents with known structures, including a Waymo L4 autonomous taxi.

We then extend this method to agents with unknown internal structures. While this involves coarser approximations, we can still provide numerical estimates for their FC. We validate this black-box approach by benchmarking a range of agents. Finally, we discuss how FC instantiates several well-known theories of general consciousness and inherits their metaphysical claims.

Self-Models

Thomas Metzinger introduced the "self-model" as a single, unified construct [16]. We take a more granular approach: treating consciousness as an aggregate of specific self-models. This aligns with Graziano's Attention Schema Theory [7] and LeCun's World Models [8]. Building on earlier proposals [2], we provide a more formal grounding:

Definition 2 (Self-Model). A self-model is any internal representation that functionally models the system's own states, processes, or capacities, and whose content is available to global reasoning processes.

Consider the following statement as an intuitive indicator:

> "I can run faster than you."

This phrase implies the agent possesses an internal representation of its physical capabilities. To produce such a comparison, the underlying model must exhibit several observable characteristics:

- **Domain:** The functional area covered (e.g., kinematics).
- **Conceptual Breadth (B):** The range of features tracked (e.g., "running" and "jumping" performance).
- **Conceptual Depth (D):** The granularity of information (e.g., statistics on peak velocity or endurance).

– **Reasoning Power (P):** The performance of the inference engine operating on the model to predict the future (e.g., extrapolating performance to a future race).

Quantifying these properties can be difficult. Domains often overlap; a "spatial" model may be inseparable from a "body" model in a vacuuming robot. Furthermore, a spatial self-model may be hard to distinguish from a first-order model of the 3D environment.

Note the absence of "interconnection" between self-models as proposed by Integrated Information Theory [23]. Instead, self-models are "illuminated" by attention, making their contents available to global reasoning processes which effectively perform the integration.

Scoring Self-Models

Our scoring system rewards agents with broad, detailed, and actionable self-representations. A "better" self-model tracks more phenomena (breadth) with higher granularity (depth), and enables the system to compute useful conclusions (reasoning power).

We define the Functional Consciousness Score (FCS) of a single self-model as the product of its representational capacity and its reasoning power. This multiplicative structure explains why legacy AI systems often exhibited low FC despite possessing detailed data: their reasoning engines were too weak to extract significant utility from the model. Conversely, a powerful reasoning engine (like a stateless LLM) scores zero if it lacks a persistent self-model to operate upon.

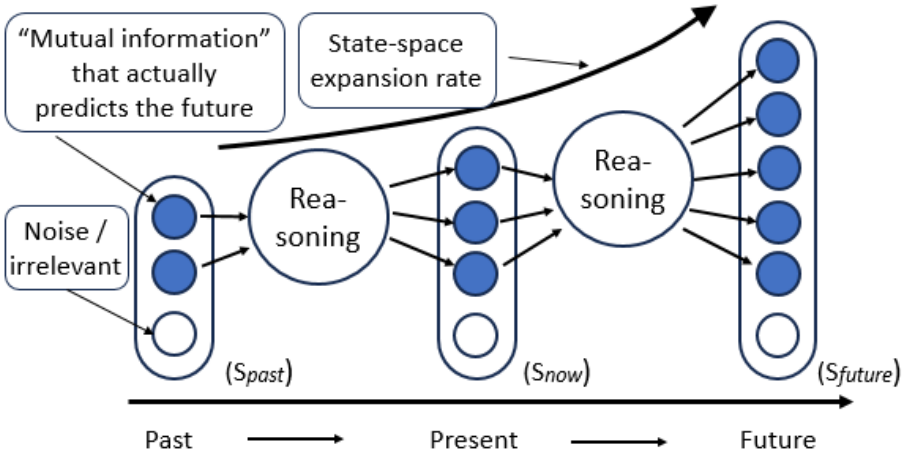

Figure 1. Scheme of State-Space Expansion and Mutual Information.

To formalize this, we draw on the language of predictive information [5] to express representational capacity (R) and reasoning power (P).

Bialek et al. argue that most information a system processes is actually useless noise. True complexity lies in the **predictive information**—the subset of mutual information from the past that actually helps predict the future.

Definition 3 (Functional Consciousness Score, FCS). For a self-model m in domain f, the Functional Consciousness Score is defined as:

$$FCS(m) = R(m) \cdot P(m) \quad (1)$$

where:

- $R(m)$ is the **Representational Capacity**, the sum of mutual information $I(m_i; s_i)$ between each independent self-model variable m_i and the corresponding system state variable s_i.
- $P(m)$ is the **Reasoning Power**, measured as the **state-space expansion rate** per reasoning cycle—the rate at which distinguishable, reachable states grow through inference over time.

Representational capacity $R(m)$ can be decomposed into two factors:

$$R(m) = B(m) \cdot \bar{D}(m) \quad (2)$$

where:

- $B(m)$ **(Breadth)** is the effective dimensionality: the number of variables tracked above a noise threshold ϵ.
- $\bar{D}(m)$ **(Depth)** is the average mutual information per tracked variable: $R(m)/B(m)$.

To summarize, R measures the state-space size and P measures the expansion rate, so $R \cdot P$ calculates the total "volume" of the conclusion manifold per reasoning cycle.

Combining Multiple Self-Models

To score an entire agent possessing multiple self-models, we must consider how its global reasoning integrates them:

No cross-reasoning: Without inter-model reasoning, the agent's total score simply sums the individual scores: $FCS_{agent} = \sum_j FCS(m_j)$.

Perfect cross-reasoning At the other extreme, an agent capable of drawing all possible conclusions across its self-models achieves "informational closure." In this case, the total capacity is additive: $R_{agent} = \sum_j R(m_j)$ The reasoning power becomes exponential with the number of combined self-models: $P_{agent} = \prod_j P(m_j)$. This aligns with the highest levels of Φ in Integrated Information Theory [23], where the system exists as a single, irreducible causal entity.

Practical cross-reasoning Real-world agents vary widely in how many state variables can be combined by global reasoning into new states. For an example of a white-box analysis of cross-reasoning in the Waymo L4 taxi please see the companion repository [3].

However, there are practical issues to describe this middle ground. Modern LLMs outperform legacy symbolic AI systems by orders of magnitude in this cross-domain integration. Techniques like abstraction and efficient heuristics expand the accessible state-space and thus directly increase the FCS_{agent}. Some self-models may be substantially more "productive" than others, either by the number of connections of the importance of the conclusions to be drawn.

Future research could formalize **Integration Density** as a graph metric: viewing self-models as nodes and the active, causal inferences between them as edges. The density of that graph literally represents the architectural bridging of the system's functional consciousness.

Scoring the Waymo L4 Spatial/Kinematic Self-Model

We illustrate the FCS metric by estimating the score of the spatial/kinematic self-models of the white-box Waymo L4 autonomous taxi. While Waymo has published parts of their architecture [24], calculating their exact internal state requires educated deductions about self-model boundaries, variable counts, simulation depth, and signal-to-noise ratios. These assumptions give our final score a wide confidence interval of roughly $\pm$ an order of magnitude.

Even so, this is a grounded first attempt at quantifying functional consciousness on actual hardware. A Waymo system engineer could significantly tighten these estimates.

We estimate $R_{waymo} \approx 560$ bits by multiplying ~40 self-model state variables (B_{waymo}) by an average depth (D_{waymo}) of ~14 bits per variable.

The variables include:

- **Kinematic Self-Model:** 12 variables (Position x, y, z; Velocity v_x, v_y, v_z; Acceleration a_x, a_y, a_z; Angular rates $\omega_{pitch,yaw,roll}$).
- **Actuator Self-Model:** 8 variables (Steering torque, braking pressure, wheel rotational speeds).
- **Task/Trajectory Self-Model:** ~20 variables (Active path spline parameters, distance-to-lane-center, time-to-arrival, collision buffers).

Reasoning Power We calculate P_{waymo} as the state-space expansion rate—the growth of the "conclusion manifold" during inference. Waymo uses Model Predictive Control (MPC) and Monte Carlo simulations for trajectory planning.

- **Initial State ($t = 0$):** 480 bits.
- **Horizon ($t = 5s$):** The system simulates thousands of branched futures based on candidate actions and actor interactions.

– **Expansion Factor:** If the inference engine generates a probability distribution over 2^{10} distinguishable and reachable future states, the expansion rate is the bit-growth of this set.

Following Bialek et al. [5], for a $K = 40$ parameter model with $N \approx 100$ inference steps per cycle:

$$P \approx \frac{40}{2} \log_2(100) \approx 20 \times 6.64 \approx 133 \text{ expansion units.}$$

As a result, we get:

$$FCS_{waymo} = R \cdot P = 560 \times 133 \approx \mathbf{74,500} \text{ FC Points} \quad (3)$$

This score indicates that while a Waymo taxi has a detailed model of its physical self and future trajectories (R), its functional consciousness is limited by narrow reasoning in its specific domain. It lacks the cross-domain "reasoning manifold" expansion seen in biological agents.

Self-Model Evaluation Comparison Table

We evaluate selected self-models of contemporary systems against two human baseline self-models.

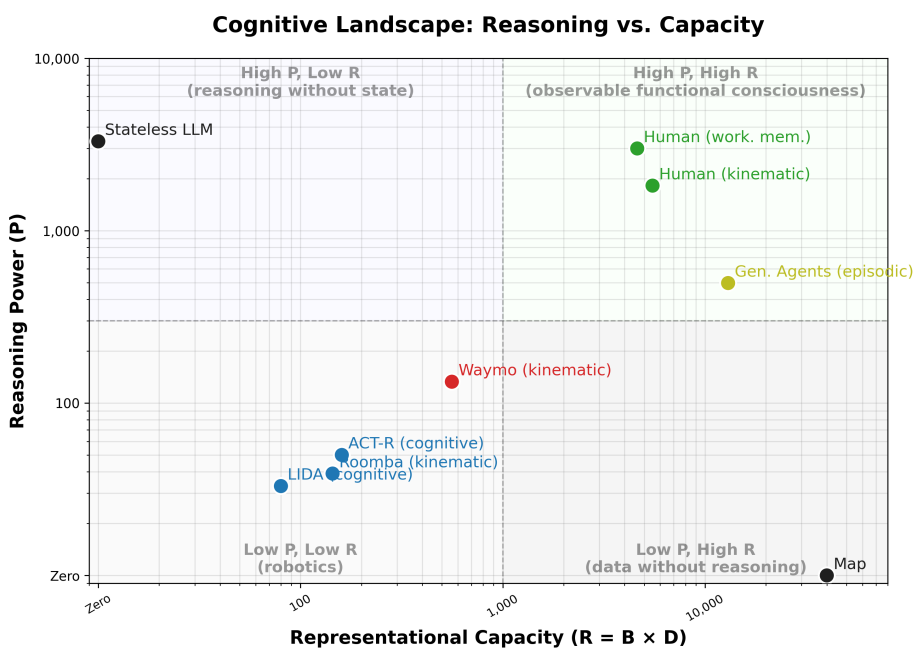

Figure 2. Reasoning Power vs. Representational Capacity (from Table 1).

This table provides short descriptions of the selected systems, the type of self-models evaluated and some of the assumptions taken. Detailed evaluation reports are available in the companion repository [11].

- **Map:** A map exhibit very rich spatial data (high R), but have no reasoning power at all (P). For evaluation we have chosen an average municipal city map with ~1k geometric shapes with ~40 bit each.
- **Stateless LLM:** LLMs (Large Language Models) act as powerful reasoning engines with performance similar to humans. We estimate $P \approx 3{,}300$ by applying Bialek scaling to the ~1,000 effectively attended tokens across ~100 Transformer layers. However, without persistent state, LLMs score zero in R and thus $FCS = 0$.
- **LIDA Cognitive Architecture:** LIDA is designed to model the "conscious cycle" via Global Workspace Theory [13]. While it possesses numerous self-models, its symbolic reasoning engine is very limited compared to modern neural approaches, leading to low R and very low P. LIDA scores below Roomba on its best single self-model. This is because LIDA's representations are shallow ($\bar{D} = 4$) and its reasoning is limited to activation arithmetic ($P = 33$). The evaluation argues this aligns with the paper's conclusion that "LIDA has been built as a consciousness show-case rather than for actual performance."
- **Roomba:** These robots use SLAM [10] to build spatial representations. They possess a basic spatial self-model but have limited reasoning power (P).
- **ACT-R:** The ACT-R cognitive architecture models reasoning through a tightly constrained central bottleneck of buffers and production rules. We evaluate its declarative memory and active buffer system (the cognitive domain). Due to its utility-learning production selection, its reasoning power ($P \approx 50$) out-scales LIDA's activation arithmetic, though its tight working memory bottleneck limits its overall capacity ($R \approx 160$).
- **Waymo L4:** The Waymo possesses sophisticated spatial self- models with integrated uncertainty, health monitoring, and trajectory simulation. It exhibits cross-domain reasoning (e.g., correlating sensor reliability with trajectory planning).
- **Generative Agents:** Stanford's "Smallville" agents [18] use a LLM with memory streams, reflection, and social interaction. They possess rich episodic and social self-models but lack embodiment. We score the `episodic` self-model, which includes the memory stream, reflection nodes, and retrieval system. Please not that the P score is lower than the Stateless LLM, because the agent architecture acts as a bottleneck.
- **Human (kinematic):** We score the narrowly defined kinematic self-model, which literature estimates to have ~550 state variables (joint angles, actuator feedback, vestibular data). The reasoning power ($P \approx 1{,}826$) reflects the cerebellum's role as a predictive forward model for motor planning.
- **Human (working mem.):** Quantifying the human episodic/cognitive self-model requires strict boundary assumptions to remain tractable. Rather than scoring the entire lifetime store of autobiographical memory, we analyze the

active working set engaged during a single episode of reflective reasoning. Drawing on cognitive science estimates of working memory and narrative reconstruction [9], we estimate $B \approx 330$ actively maintained variables (perceptual features, social actors, causal links), with $\bar{D} \approx 14$ bits reflecting rich multi-modal content. The resulting reasoning power ($P \approx 3{,}000$) reflects the massive parallelism of biological associative expansion and is comparable to a Transformer LLM ($P \approx 3{,}300$). This aligns with the empirical observation that LLMs approach human-level reasoning depth.

In the following table summarizes the benchmark results. The specific valuations are intended as order-of-magnitude estimates to illustrate the discriminatory power of the FCS metric. Rather than treating these numbers as definitive constants, the primary contribution here is the metric framework, which allows domain experts to derive precise measurements based on exact architectural details. Details for each evaluation are available in the companion repository [3].

Table 1. System Self-Model Comparison Table

System	B (v)	$\bar{D}$ (bits/v)	P	FCS
Map	~1k	~40	0	0
Stateless LLM	0	0	~3,300	0
LIDA (cognitive)	~20	~4	~33	~2.6k
Roomba (kinematic)	~18	~8	~39	~5.6k
ACT-R (cognitive)	~20	~8	~50	~8.0k
Waymo (kinematic)	~40	~14	~133	~74.5k
Gen. Agents (episodic)	~130	~100	~497	~6.5M
Human (kinematic)	~550	~10	~1,826	~10M
Human (working mem.)	~330	~14	~3,000	~13.9M

While some AI systems match human performance in narrow domains (e.g., an LLM's raw reasoning power), they rarely combine high capacity and reasoning within a single self-model. Furthermore, the human mind integrates many such models across domains, leading to an exponential growth with the number of models.

Functional Self-Model Analysis (FSMA)

We could effectively conclude the paper here, having established that self-models serve as measurable functional units for AGI evaluation in white-box systems. However, to fulfill the promise of a universal metric, we must be able to benchmark complex black-box systems where internal variables are hidden. Without a behavioral inference tool, our core claim—that functional consciousness can be practically scored—would break down for advanced agents like LLMs or humans.

To bridge this gap, we introduce Functional Self-Model Analysis (FSMA), an abductive behavioral inference method that detects the presence and richness

of self-models solely through the predictive structure of observed outputs. The intuition is straightforward:

> "If an agent can repeatedly and consistently talk about a state, it must possess a functional model of that state."

This is analogous to functional data dependency [8] and abductive reasoning—inferring the necessary preconditions for an observed outcome.

The VAT Intuition To illustrate this, consider an automated accounting system that calculates Value Added Tax (VAT) for international transactions. By law, the VAT rate depends on:

- The origin of the item.
- The destination of the item.
- The category of the item (e.g., food vs. services).

If the system consistently identifies the correct VAT across diverse invoices, we can abductively conclude that:

1. The system possesses data representing the origin, destination, and category.
2. The system possesses a functional mapping VAT $= f$(origin, destination, category).

Whether this mapping is a lookup table, a hard-coded rule, or a latent neural vector is irrelevant to the functional analysis; its presence is a prerequisite for the observed behavior.

Definition 4 (Functional Self-Model Analysis, FSMA). FSMA identifies the minimal set of self-models $M(f)$ required for a consistently produced observable output f:

$$M(f) = \{m_i \mid m_i \text{ is a minimal self-model required for } f\} \quad (4)$$

Each self-model m_i is defined as a tuple $\langle D_i, S_i, O_i \rangle$, where:

- D_i **(Data Domains):** The specific areas represented (e.g., spatial, temporal).
- S_i **(Structural Form):** The architecture of the representation (e.g., 3D scene graph, directed acyclic graph, attention map).
- O_i **(Operations):** The functional capabilities enabled (e.g., simulation, prediction, counterfactual reflection).

"Minimal" refers to the model with the lowest representational capacity required to satisfy the observation. Identifying the absolute minimum helps decide between multiple candidate solutions. For example, consider this reflection from Virginia Woolf's *The Mark on the Wall*:

> "All the time I'm dressing up the figure of myself in my own mind, lovingly, stealthily. . . "

To generate this thought, an agent must possess an active `episody-narrative` self-model, but more importantly, it requires a concurrent `meta-self-awareness` model. The system is explicitly monitoring its own ongoing attempt to rewrite its internal parameters to optimize for positive affect (ego). The absolute minimal requirement to output this sentence is a dual-layered architecture: a primary model actively constructing a narrative identity, and a secondary attention mechanism observing that construction process in real-time.

Empirical Evaluation of FSMA

To test FSMA, we needed datasets rich in internal state references. While "Descriptive Experience Sampling" (DES) [18] catalogs human inner experience, many existing sets use restrictive categories. We instead turned to early 20th-century stream-of-consciousness literature. Though crafted for aesthetics, these texts offer high-density subjective flows. The core logic of FSMA still holds against the "parroting" or "word-to-world" fallacies: to repeatedly and consistently describe thoughts and perceptions as these authors do, the narrator must possess sophisticated underlying self-models. We selected Virginia Woolf's "The Mark on the Wall" [25] for its manageable length and dense meta-cognitive reflection. The annotated text is included in the companion repository [3].

The Self-Model Catalog During our FSMA of "The Mark on the Wall", we extracted all self-referential expressions to identify candidates for the "minimal" self-models required to generate them. This process yielded a comprehensive Self-Model Catalog.

Through this fieldwork, we encountered the potential fallacies of Schleiermacher's hermeneutic circle [22], which we navigated using two distinct modes of analysis:

1. **Bottom-Up Analysis:** An inductive process extracting self-model candidates directly from the text, grouped by shared data requirements.
2. **Top-Down Analysis:** A deductive validation of existing theoretical frameworks (e.g., a specific cognitive architecture) against the models identified in the text.

Bottom-up analysis reveals the inherent complexity of self- representation. For example, we initially identified overlapping models like a "dual self" (private vs. public) and a "multi-faceted self" (context-dependent identity). While philosophically interesting, these often function as special cases of broader "Theory of Mind" (ToM) or "Narrative Self" models. To maintain a lean, benchmark-ready catalog, we aggregate these into general categories based on functional similarity.

In contrast, top-down analysis carries a significant methodological risk: if the analysis begins with a predefined architecture, the investigator may primarily detect only the models that the framework already assumes. This can lead to confirmation bias.

We address this dilemma by decoupling the FSMA methodology from any single catalog. This allows researchers to apply their own theoretical frameworks, leaving scientific discourse to determine which catalogs offer the most predictive power for AGI.

The SBR-Catalog of Self-Models The following catalog results from a bottom-up FSMA of Woolf's text, filtered through the lens of the Scene-Based Reasoning (SBR) cognitive architecture [4]. SBR details several first-order subsystems and assumes an attention subsystem [12] that can focus on both external objects and internal self-models.

The SBR-derived catalog consists of 46 self-models across ten functional areas (see the companion repository [3]):

- Body (5): `body-3d`, `body-kinematic`, `body-sensor`, `body-actuator`, `body-energy`.
- Spatial (4): `spat-relative`, `spat-trajectory`, `spat-collision`, `spat-tool`.
- Action/Planning (5): `action-tree`, `action-perform`, `action-progress`, `action-plan`, `action-improv`.
- Goal/Motivation (3): `goal-tree`, `goal-reward`, `goal-conflict`.
- Cognitive (5): `episody`, `episody-narrative`, `episody-time`, `mem-avail`, `learn-rate`.
- Informational/Knowledge (7): `inf-know`, `inf-fresh`, `inf-creative`, `inf-consistency`, `inf-reasoning`, `inf-hypo`, `inf-confidence`.
- Emotional/Affective (4): `mood`, `mood-needs`, `mood-stress`, `mood-load`.
- Social/Interaction (6): `social-tom`, `social-role`, `social-comm-state`, `social-trust`, `social-influence`, `social-empathy`.
- Meta/Reflexive (4): `meta-attention`, `meta-self-awareness`, `meta-explain`, `meta-accuracy`.
- Ethics/Safety (3): `ethics`, `ethics-safety`, `ethics-drift`.

Perception is excluded from this catalog as a first-order activity with several self-models monitoring the perceptual process (e.g., `body-sensor` for sensory limits). "Awareness" of perception is thus modeled as attention toward these self-models.

Benchmarking Agents with the SBR Self-Model Catalog

While previous sections focused on the numerical FC score of white-box agents with limited capabilities and detailed internal information available, here we benchmark biological and more complex AI systems. Clearly, the precision of our evaluation will degrade even further, which will be reflected in the way we will visualize the results. Still, this is a major improvement to the previously available philosophical models, as it allows discourse using scientific tools. Again, these results are presented to illustrate the discriminatory power of the FC score metric.

Visualizing Cognitive Shape In the previous section, we derived a catalog of 46 self-models across ten functional areas using Functional Self-Model Analysis (FSMA) on literature. We can now use this catalog to benchmark the same selection of AI agents and human baselines from Table 1, providing a comprehensive view of their self-models across all domains.

Methodological Boundary: It is important to explicitly distinguish between the two methodologies used here. For white-box agents (e.g., Waymo, Roomba, ACT-R), we calculate the presence of self-models directly from their architectural specifications. However, for black-box systems (e.g., Generative Agents, LLMs, and humans), we must rely on functional self-model analysis, using behavioral abduction to infer the minimal self-models required to produce their observable outputs. Despite the different methods, both approaches estimate the same underlying property: the informational richness of the agent's internal state modeling and the associated reasoning power P.

Extending the precise white-box FCS math ($R = B \cdot \bar{D}$) across all ten domains for every agent is impractical. Instead, we use a rapid approximation: mapping the presence and richness of self-models relative to a human baseline (normalized to 100%). Plotting these values on a radar chart reveals an agent's **cognitive shape**. These charts represent structural breadth rather than absolute, rigorous scores.

The Role of Global Reasoning (P) While the cognitive shape reveals *where* an agent possesses self-models, its total functional consciousness depends heavily on the *reasoning power* (P) operating over those models.

As discussed earlier, P determines the scale of the state-space expansion. Even if local reasoning differs slightly across modules, the global reasoning engine acts as an exponential multiplier. This is visually demonstrated when comparing radar charts and FCS scores: two agents might possess a similar, broad cognitive shape (e.g., LIDA and Generative Agents both touch upon many domains), but because they operate at radically different P levels (legacy symbolic arithmetic vs. massively parallel neural inference), their resulting FC scores differ by three orders of magnitude. The Shape shows the potential; the P-scale determines the actualized utility. For the detailed evaluation reports please see the companion repository [3].

Relation with Other Theories of Consciousness

Functional Consciousness (FC) was not proposed as a competing metaphysical account of phenomenal experience or the "hard problem." However, FC instantiates and details several major theories, inheriting their claims and serving as a bridge between abstract philosophy and concrete engineering.

Fig. 1. Figure 3. FSMA radar charts across 8 representative agent architectures. Each chart shows the qualitative "cognitive shape" (filled area) against the human baseline.

Integrated Information Theory Giulio Tononi's Integrated Information Theory (IIT) identifies consciousness with integrated information (Φ) [23]. While often criticized as computationally intractable, we can map its core ideas onto engineering metrics: IIT's "differentiation" maps to our representational capacity (R), and its "integration" maps to our reasoning power (P).

It is even possible to calculate an integration measure similar to Φ of an agent where subsystems are governed by multiple, overlapping self-models. Total integration can be viewed as the difference between collective reasoning power and the sum of the reasoning power of partitioned modules. We can approximate this by measuring how much reasoning power (P) is lost when we disconnect self-models from their underlying subsystems:

$$\Phi_{FCS} = P(S) - \sum_j P(module_j) \quad (5)$$

Where:

- $P(S)$: The total reasoning power of the entire integrated system.
- $module_j$: The j-th partitioned subsystem combined with its dedicated, non-shared self-models.

In summary, FC decomposes Φ into measurable components and operationalizes "interconnection" using a standard apparatus for reasoning and attention [12].

Higher-Order Theories Higher-Order Theories (HOT/HOP) [21] argue a mental state is only conscious when it becomes the target of a higher-order state. Functional Consciousness maps directly to this: self-models (m_i) act as higher-order representations monitoring first-order states (s_i).

A state s_i becomes an object of access consciousness when the system's reasoning engine (P) processes an associated self-model (m_i). The respective self-model is "illuminated" by attention and becomes available to the global reasoning processes. Recursion—such as the `meta-attention` model monitoring attention itself—creates the **meta-reflective cascades** characteristic of advanced cognition.

Workspace and Predictive Theories Global Workspace Theory (GWT) [1] suggests that consciousness arises when information is "broadcast" to a wider network of cognitive processes. FC operationalizes this broadcasting via reasoning across self-models.

Predictive Processing (PP) describes the brain as a prediction engine. Our use of predictive information [5] to measure representational capacity (R) grounds functional consciousness firmly in the PP tradition. In this view, consciousness is the process of minimizing the prediction error of the agent's own internal self-models.

Summary Position Many leading theories of consciousness implicitly rely on some form of self-representation, global integration, or meta-reflection. Functional Consciousness makes these structural properties explicit and quantifiable. In this sense, FC serves as an implementation layer that instantiates and unites these diverse theoretical accounts.

Conclusions

Returning to the title, we have shown that Functional Consciousness instantiates the core structural claims of several prominent consciousness theories while remaining empirically grounded and tractable to compute. By moving beyond the "hard problem," it provides engineers with a behavioral metric for evaluating how well an agent models and reasons about its own internal states.

Our results demonstrate that FC can successfully benchmark a diverse range of systems—from autonomous taxis to 20th-century literature. We show that:

1. **Self-Models:** Self-models are a useful new unit of analysis for AGI evaluation.
2. **Metric Clarity:** The $FCS = R \cdot P$ formula captures the essential trade-off between representational capacity and reasoning utility.

3. **Integration Value:** The exponential growth of cross-domain reasoning explains the significant leap from narrow AI to general biological consciousness.
4. **Benchmarking Utility:** FSMA provides a systematic way to extract and compare self-models from behavioral output, even in black-box systems.

Outlook for AGI As we advance toward Artificial General Intelligence (AGI), functional consciousness offers a crucial safety and evaluation tool. An agent that cannot accurately reason about its own capabilities, goals, and limitations is inherently unreliable. By measuring and optimizing for FC, we can engineer systems that are not only capable, but structurally transparent and self-aware.

To increase FC in future systems, we recommend building AI with **inspectability** in every subsystem and exposing the results to global reasoning via an attention filter.

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